

Mathematical Background and Implementation Guide

N2SCFTDB / n2_theory_index.py

FORM and LiE computation of four-dimensional $\mathcal{N} = 2$ indices
Simple and semisimple product gauge groups

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Purpose. The canonical implementation computes the flavor-unrefined right superconformal index of a Lagrangian four-dimensional $\mathcal{N} = 2$ gauge theory. FORM performs exact truncated series algebra. LiE decomposes individual Adams operations and intermediate products. Python pairs dual irreps to extract the final singlet, independently for every simple gauge factor. SQLite stores exact parsed FORM expansions, decompositions and final singlet coefficients. Sage supplies input/representation data, the dual-label convention and the exact returned polynomial ring.

This revision adds FORM/TFORM execution controls, CPU allocation and the dated 32-worker benchmarks; the public sector API is documented as well. Earlier mathematical and cache material is retained. GUI/search/export and production schema 1 belong to the companion implementation reference. Preserved pages 2–4 contain the original setup and baseline recurrence; Section 5.4 gives the current generated code.

Contents

1 Scope and program architecture	2
2 Index convention and single-letter functions	2
3 Matter characters for simple and product groups	3
4 Plethystic exponential and exact cutoffs	3
5 How FORM represents and expands the index	3
Current degree-bounded loop, proof and example	5
TFORM controls and 32-worker measurements	8
6 Adams operations and LiE decomposition	11
7 Gauge-singlet projection for product groups	12
Disconnected sectors and stored-index reuse	14
8 SQLite caches and complete Adams-cache builder	17
9 Python functions and their mathematical roles	22
10 JSON examples and command-line use	23
11 Validation and representative performance	24
12 Interpretation and limitations	28
References	29

1 Scope and program architecture

The supported gauge algebra is a finite semisimple algebra

$$\mathfrak{g} = \bigoplus_{a=1}^s \mathfrak{g}_a, \quad G = \prod_{a=1}^s G_a, \quad (1)$$

where every $\mathfrak{g}_a$ has Cartan type $A_r, B_r, C_r, D_r, E_6, E_7, E_8, F_4$, or G_2 . The compact group convention is the simply connected one used by the shared Lie-algebra module; in particular C_r is written $\mathrm{Sp}(r)$. Abelian gauge factors are outside the current input model.

The calculation has four layers:

1. `check_input_data` validates the existing simple- or product-group JSON convention and resolves names to Dynkin labels.
2. `_build_form_program` constructs the truncated plethystic exponential with formal character symbols.
3. `CharacterDecompositionCache` asks LiE for the necessary virtual character decompositions and caches decompositions and singlet coefficients in SQLite.
4. `_project_terms` extracts the gauge singlet factor by factor and `_to_sage_polynomial` packages the exact answer.

This division is important: FORM never needs to know a root system, while LiE never needs to manipulate the fugacities t, y, u .

2 Index convention and single-letter functions

Choose the supercharge defining the right index. With the conventions of Gadde, Pomoni, and Rastelli, the BPS condition and trace are [1]

$$\delta_R = \Delta - 2\bar{j} - 2R + r = 0, \quad (2)$$

$$\mathcal{I}(t, y, v) = \mathrm{Tr}_{\delta_R=0} (-1)^F t^{2(\Delta+\bar{j})} y^{2j} v^{-r-R}. \quad (3)$$

The code substitutes

$$v = u^2, \quad (4)$$

so every computational power of u is integral.

The vector and half-hypermultiplet single-letter indices are [1]

$$f_V(t, y, u) = \frac{t^2 u^2 - t^4 u^{-2} - t^3(y + y^{-1}) + 2t^6}{(1 - t^3 y)(1 - t^3 y^{-1})}, \quad (5)$$

$$f_H(t, y, u) = \frac{t^2 u^{-1} - t^4 u}{(1 - t^3 y)(1 - t^3 y^{-1})}. \quad (6)$$

The denominator counts the two compatible derivatives:

$$\frac{1}{(1 - t^3 y)(1 - t^3 y^{-1})} = \sum_{p, q \geq 0} t^{3(p+q)} y^{p-q}. \quad (7)$$

FORM uses a finite rectangular sum with $0 \leq p, q \leq \lfloor D/3 \rfloor$. Terms whose combined t degree exceeds D are removed automatically.

3 Matter characters for simple and product groups

For a product group, an irreducible external tensor product is

$$\mathbf{R}_h = \boxtimes_{a=1}^s R_{h,a}, \quad \chi_{\mathbf{R}_h}(g_1, \dots, g_s) = \prod_{a=1}^s \chi_{R_{h,a}}(g_a). \quad (8)$$

An omitted representation in the JSON input is the singlet, whose character is one.

The adjoint representation of the product algebra is a direct sum,

$$\text{ad}_{\mathfrak{g}} = \bigoplus_{a=1}^s (\mathbf{1}, \dots, \text{ad}_{\mathfrak{g}_a}, \dots, \mathbf{1}), \quad \chi_{\text{ad}_{\mathfrak{g}}} = \sum_{a=1}^s \chi_{\text{ad}_a}. \quad (9)$$

This explains why vector-multiplet characters are added, whereas a bifundamental hypermultiplet character is multiplied across its gauge factors.

For multiplicity N_h , the character measured in half-hyper units is

$$\hat{\chi}_h = \begin{cases} N_h \chi_{\mathbf{R}_h}, & \text{half hyper,} \\ N_h (\chi_{\mathbf{R}_h} + \chi_{\overline{\mathbf{R}}_h}), & \text{full hyper.} \end{cases} \quad (10)$$

Conjugation is simultaneous and factorwise:

$$\overline{\mathbf{R}}_h = \boxtimes_{a=1}^s \overline{R}_{h,a}. \quad (11)$$

Thus a full hyper in $R_1 \boxtimes R_2$ contains $R_1 \boxtimes R_2$ and $\overline{R}_1 \boxtimes \overline{R}_2$; it does not contain the two mixed tensor products unless they are separate matter entries.

4 Plethystic exponential and exact cutoffs

With flavor fugacities unrefined, define the representation-valued letter index

$$\mathcal{L}(t, y, u; g) = f_V(t, y, u) \sum_{a=1}^s \chi_{\text{ad}_a}(g_a) + f_H(t, y, u) \sum_h \hat{\chi}_h(g). \quad (12)$$

The free-field index before gauge projection is the plethystic exponential [1, 2]

$$\mathcal{Z}(t, y, u; g) = \text{PE}[\mathcal{L}] = \exp E(t, y, u; g), \quad (13)$$

$$E(t, y, u; g) = \sum_{j=1}^{\infty} \frac{1}{j} \mathcal{L}(t^j, y^j, u^j; g^j). \quad (14)$$

Since every letter begins at t^2 , a result through t^D needs only

$$1 \leq j \leq M = \left\lfloor \frac{D}{2} \right\rfloor. \quad (15)$$

The same lower bound implies that only $E^k/k!$ with $k \leq M$ can contribute. Consequently both the Adams sum and the ordinary exponential have finite, exact cutoffs at the requested order.

5 How FORM represents and expands the index

5.1 Formal character symbols

Each distinct pair “gauge-factor position, Dynkin labels” receives a commuting FORM function $C_p(j)$. Mathematically it denotes the j -th Adams operation

$$C_p(j) = \psi^j(\chi_{R_p}), \quad (\psi^j \chi)(g) = \chi(g^j). \quad (16)$$

For a bifundamental, FORM therefore sees $C_p(j)C_q(j)$. It expands only the coefficients and products of these formal objects; their Lie-theoretic decomposition is postponed until after the series is complete.

At Adams degree j , `_build_form_program` constructs

$$\mathcal{L}_j = f_V(t^j, y^j, u^j) \sum_{p \in \mathcal{V}} C_p(j) + f_H(t^j, y^j, u^j) \sum_m N_m \prod_{p \in m} C_p(j), \quad (17)$$

where $\mathcal{V}$ contains the adjoint character of every gauge factor and the matter monomials already include the full-hyper conjugate term from Eq. (10).

5.2 The generated FORM program in plain language

The central generated block has the following form:

Listing 1: Schematic FORM program generated by Python.

```
S m,n,y,idx1,idx2,j,z,u,t(:D);
CF d,C0,C1,...;
PolyRatFun d;
Function Kvec,Khyp;

L J=sum_(idx1,0,D/3,m^idx1)
    *sum_(idx2,0,D/3,n^idx2);
id m=t^3*y;
id n=t^3/y;
.sort

L itotal=sum_(j,1,D/2,(vector terms + hyper terms)/j);
id Kvec(t?,y?,u?)=J*(t^2*u^2-t^4/u^2-t^3*y-t^3/y+2*t^6);
id Khyp(t?,y?,u?)=J*(t^2/u-t^4*u);
.sort
```

FORM construct	Meaning in this calculation
S ... t(:D);	Declares commuting symbols. The maximum-power declaration automatically discards $t^{D+1}, t^{D+2}, \dots$ during term normalization [4, Sec. 7.143].
CF d,C0,...;	Declares commuting functions. The $C_p(j)$ objects are formal Adams-operated characters.
PolyRatFun d;	Makes <code>d(numerator,denominator)</code> the exact rational coefficient container, so equal monomials combine without floating point arithmetic [4, Sec. 7.114].
L name=...;	Creates a local FORM expression. <code>J</code> , <code>itotal</code> , <code>I</code> , and <code>result</code> are successive exact expressions.
sum_(j,a,b,X)	Generates the finite sum $\sum_{j=a}^b X$; the last argument is the summand [4, Sec. 8.67].
id lhs=rhs;	Applies an algebraic substitution. A question mark denotes a wildcard, so <code>Kvec(t?,y?,u?)</code> matches arguments such as <code>Kvec(t^j,y^j,u^j)</code> [4, Secs. 7.70–7.71].
.sort	Ends a FORM module, expands products, combines identical terms, and enforces the t cutoff before the next module [4, Sec. 2.3].

5.3 Why the auxiliary symbol z gives the exponential

Let $E = \text{itotal}$ and $M = \lfloor D/2 \rfloor$. The code begins with $I = zE$. For $k = 2, \dots, M$ it applies

$$z \longmapsto 1 + \frac{zE}{k}. \quad (18)$$

After the k -th step, induction gives

$$I_k = \sum_{r=1}^{k-1} \frac{E^r}{r!} + z \frac{E^k}{k!}. \quad (19)$$

The final substitution $z \mapsto 1$, followed by adding the vacuum term, yields

$$1 + I_M = \sum_{r=0}^M \frac{E^r}{r!} = \exp(E) + O(t^{D+1}). \quad (20)$$

This is an ordinary exponential of a representation-valued series; Bose and Fermi statistics have already entered through the signed single-letter functions.

5.4 Degree-bounded FORM exponential: current implementation

The 16 September implementation in `_build_form_program` evaluates the same finite exponential as the auxiliary-variable recurrence above, but avoids products that cannot survive the requested cutoff. Let N denote that cutoff (called D in the earlier formulas), and write

$$A = \text{itotal} = \sum_{d=2}^N A_d t^d, \quad [P]_N = \sum_{d \leq N} [t^d] P t^d, \quad M = \lfloor N/2 \rfloor.$$

The coefficients A_d contain Laurent monomials in y, u and formal gauge characters. They contain no t . All powers of t are nonnegative and the exponent starts at degree two. The target before gauge projection is $[\exp(A)]_N$. The derivation below follows the project's generated program.

Step 1: construct the derivative series and plethystic exponent. The expression J is the truncated product of geometric series for $(1 - t^3 y)^{-1} (1 - t^3 / y)^{-1}$. Substituting the vector and hyper letters into `itotal` gives

$$A = \left[\sum_{j=1}^M \frac{1}{j} \left(f_V(t^j, y^j, u^j) \sum_{p \in V} C_p(j) + f_H(t^j, y^j, u^j) \sum_m N_m \prod_{p \in m} C_p(j) \right) \right]_N.$$

Here j is the Adams index. It differs from the outer Taylor-power index i below. Integer multiplicities N_m are unchanged by Adams substitution. The bound on j follows from $2j \leq N$; the derivative rectangle uses `order//3`. The declaration `t(:N)` remains a final degree guard.

Step 2: organize the exponent and initialize the active power.

```
Bracket t;
.sort
Skip J,itotal;
L I=z*itotal;
.sort
```

`Brackett` groups equal t powers. A reference `itotal[t^d]` returns A_d , so multiplication must restore `t^d` explicitly. `SkipJ,itotal;` preserves those expressions, including the bracket information, for the current module. They remain readable on the right-hand side. Repeat `Skip` after each module boundary; it is not a permanent freeze [11, entries “bracket”, “skip”, and Chapter 9].

Initially $I = zA$. The symbol z marks the one Taylor order still eligible to generate the next order; unmarked terms have already been accumulated. Neither z nor the temporary symbol w is a physical fugacity.

5.5 One exponential step, line by line

For example, at cutoff $N = 8$ the following is the actual emitted loop and finalization block; Python substitutes the integer bounds before running FORM:

```
#do i=2,4
  Skip J,itotal;
  #do k=2,6
    if (count(t,1) == 'k');
      id z=1+w*sum_(idx1,2,8-'k',itotal[t^idx1]*t^idx1)/'i';
    endif;
  #enddo
  id z=1;
  id w=z;
  .sort:step 'i';
#enddo

L result=1+I;
id z=1;
.sort
Print result;
.end
```

Step 3: select the current term's degree. The inner preprocessor loop creates one branch for each $k = 2, \dots, N - 2$. `count(t,1)` tests the current term's t degree. An unmarked term is unchanged because it has no z to replace. On a marked term zct^k , the substitution has the exact effect

$$zct^k \mapsto ct^k + \frac{wc}{i} \sum_{d=2}^{N-k} A_d t^{k+d}.$$

The first branch retains the old Taylor contribution. The second produces the next contribution, dividing by i to turn $(i - 1)!$ into $i!$. Its summation bound is the condition $k + d \leq N$, enforced before forming the product rather than only when an excessive power is normalized away.

Step 4: protect newly generated terms. New terms carry w , not z . A later k branch in the same outer iteration can encounter one of their new degrees, but its `idz=...` cannot expand them again. Reusing z immediately would allow a single Taylor step to multiply a newly produced term more than once.

Step 5: finalize old terms, then mark the new ones. `idz=1;` removes any surviving old marker, including terms at degrees $N - 1$ and N . Then `idw=z;` makes only the new generation active. Reversing this order would erase the new marker. Finally `.sort:step'i';` collects equal terms and ends the module; the step label is diagnostic. The next iteration reinstates `Skip` for its own module.

5.6 Why the bounds and factorials are exact

For $2 \leq i \leq M$, induction on the previous step gives

$$\begin{aligned} I_1 &= zA, \\ I_2 &= A + z[A^2/2!]_N, \\ I_3 &= A + [A^2/2!]_N + z[A^3/3!]_N, \\ I_i &= \sum_{r=1}^{i-1} [A^r/r!]_N + z[A^i/i!]_N. \end{aligned}$$

Only the last summand has a marker. Step $i + 1$ retains it without a marker and multiplies it by $A/(i + 1)$, keeping precisely the products of total degree at most N . This proves the invariant with exact rational coefficients.

Why the inner loop stops at $N - 2$. Every term of A has degree at least two, so an old term of degree k can produce a retained new term only when $k + 2 \leq N$. Terms of degree $N - 1$ and N are already present and are retained by `idz=1;`. Thus $N - 2$ is a bound on terms that can be multiplied again, not a reduced output cutoff.

Why the outer loop stops at $\lfloor N/2 \rfloor$. A^i has minimum degree $2i$. No higher power can contribute. Adding the vacuum and removing the last marker therefore gives

$$\text{result} = 1 + \sum_{r=1}^M [A^r/r!]_N = [\exp(A)]_N.$$

The public API returns 1 immediately for orders zero and one. Orders two and three need only $1 + A$; Python omits the multiplication loop. These bounds use the full-index grading; a Hall–Littlewood letter beginning at degree one would require different bounds.

Worked example, $N = 6$. Set $A = at^2 + bt^3$, with commuting exact coefficients that may themselves be formal characters. Then

$$\begin{aligned} I_2 &= at^2 + bt^3 + z \left(\frac{a^2}{2}t^4 + abt^5 + \frac{b^2}{2}t^6 \right), \\ I_3 &= at^2 + bt^3 + \frac{a^2}{2}t^4 + abt^5 + \frac{b^2}{2}t^6 + z \frac{a^3}{6}t^6. \end{aligned}$$

In step three, only the marked degree-four term can multiply again. The degree-five and degree-six contributions become unmarked without being lost. The final coefficient of t^6 is $b^2/2 + a^3/6$, as in the ordinary exponential.

Validation and cache effect. `test/test_form_degree_bound.py` checks 42 low-order formal expansions (seven cases at six cutoffs) and one connected bifundamental at order 18 against the former multiplication. The 16 September recorded full run passed 357 tests in 64.117 s, with no skips. A matched, sequential FORM-only comparison at order 18 took 2.236 s before and 0.585 s afterward (3.82 times faster), with exactly the same 28,154 formal terms. This is one measurement, not a universal speedup. Evidence is retained in `output/benchmarks/degree_bounded_form_20260916.json`. The changed raw program has a new FORM-cache key. Existing rows remain valid for their original programs, and character-cache entries remain reusable.

5.7 FORM or TFORM: execution controls and CPU allocation

The symbolic program and exact projection algorithm are unchanged by choosing TFORM. `run_form` in `common/form_utils.py` validates `form_threads` as an exact positive integer. With one thread it invokes ordinary FORM; above one it invokes the separate TFORM executable:

```
form -q program.frm
tform -w32 -q program.frm
```

Both run in a temporary working directory. Missing executables, timeout, nonzero exit status or nonempty stderr are reported as errors. The production wrapper captures complete stdout in Python; selecting TFORM does not make parsing, projection or database writes threaded.

Full-index and Coulomb-index/PE/PL entry points propagate `form_threads=1` and `tform_executable="tform"`. The full-index and database-index CLIs expose `--form-threads` and `--tform-executable`, alongside the existing FORM path and timeout. For example, from the project root:

```
sage -python -m index.n2_theory_index theory.json \
--order 18 --form-threads 32 --tform-executable tform \
--processes 1 --char-cache-database /tmp/characters.db \
--form-cache-database /tmp/expansions.db
```

Use a new empty FORM cache when measuring an expansion. The raw program is the cache key and does not include the engine, its version or thread count. A hit can therefore reuse a prior engine's parsed result without running TFORM. Stored-sector reuse can bypass the expansion too.

Two levels of parallelism. The index coordinator uses the saved CPU budget P and FORM worker count T together. For $C > 0$ selected theories:

$$W_{\text{index}} = \min(C, \max(1, \lfloor P/T \rfloor)).$$

Here CPU cores `-1` resolves to the logical CPU count, or one if unavailable. On a 32-CPU host, $T = 1, 8, 32$ gives at most 32, 4, 1 simultaneous theory workers. If $T > P$, one theory still runs with the requested T threads; the setting does not silently clamp TFORM. Each theory worker keeps inner LiE/character work at one process. Anomaly enumeration and CSV export use their own CPU allocation rules. These are implementation contracts, not mathematical identities.

Memory and scope. A single expansion with many TFORM workers can still consume several GiB. Requested workers and observed OS thread counts differ; the recorded 32-worker runs reached up to 63 OS threads. The measured RSS is the complete TFORM process, excluding the Sage parent, page cache and later parsing/projection. The controls allow concentrating CPU work on one index; they do not establish a RAM bound.

Recorded checks, 16 September. Connected/disconnected full indices, fractional Coulomb PE/PL, engine-independent cache reuse, settings reload and worker allocation were checked in the implementation run: 340 passed, 38 skipped. See `test/test_form_threads.py`. The measurements on the next pages are separate one-run-per-theory experiments, not those regression tests.

5.8 TFORM benchmark: 16 repeated timeout failures

Measured 16 September 2026 on an AMD Ryzen 9 7950X (32 logical CPUs, 30.5 GiB RAM), with TFORM 5.0.0. The user selected the 16 cases that timed out again in `log_index_20260916_131926_021494.log`. All have one connected gauge sector. Each case ran separately through t^{18} using the production degree-bounded generator, `tform-w32`, and a 600-second expansion limit.

GNU time measured startup, expansion and complete text output to a local file. Compression happened afterward. These times exclude Sage preparation, output parsing, LiE projection, final polynomial construction, cache operations and MySQL writes. No existing database or cache was modified.

Theory ID	Gauge group	TFORM (s)	RSS (GiB)	Retry timeout (s)
20036	$SU(4) \times SU(5)$	24.32	7.19	122.86
20037	$SU(4) \times SU(5)$	24.45	7.19	122.88
20150	$SU(4) \times SU(6)$	23.94	7.19	123.52
20168	$SU(4) \times SU(6)$	24.32	7.19	121.50
20517	$SU(5) \times SU(5)$	37.69	8.84	121.61
20535	$SU(5) \times SU(5)$	38.33	8.84	121.32
20631	$SU(5) \times SU(6)$	37.79	8.84	121.91
20632	$SU(5) \times SU(6)$	37.58	8.84	122.61
20641	$SU(5) \times SU(6)$	24.64	7.18	122.79
20643	$SU(5) \times SU(6)$	24.92	7.19	122.06
20651	$SU(5) \times SU(6)$	25.46	7.26	122.20
20658	$SU(5) \times SU(6)$	25.47	7.25	122.50
20668	$SU(5) \times SU(6)$	24.76	7.18	121.60
20671	$SU(5) \times SU(6)$	24.74	7.18	121.63
22470	$SU(11) \times SU(11)$	37.55	8.84	122.87
22475	$SU(11) \times SU(11)$	37.46	8.83	122.92

All 16 expansions completed successfully: 23.94–38.33 s each, median across theories 25.19 s, total 473.42 s; maximum RSS 8.84 GiB. The original log ran 32 theories concurrently. Its start-to-error intervals are censored and include preparation/scheduling; their quotient with isolated TFORM time is not a controlled FORM speedup.

Success required zero exit status, empty stderr and complete printed output. Two pairs with identical raw programs had identical complete outputs (20517/22475 and 20535/22470). Every large case was not compared against serial FORM, and no complete gauge projection was benchmarked here.

Evidence. `output/benchmarks/tform32_repeated_timeouts_20260916/` contains `report.md`, `input/log_evidence`, machine-readable measurements, the benchmark script, generated programs, hashes and compressed outputs. The script overwrites its own output directory; copy it and change its output path before an independent repeat.

5.9 TFORM benchmark: random recovered cases and save-time ratios

The initial run `log_index_20260916_130305_574084.log` had 210 FORM timeouts. Later logs at 13:19:26 and 13:52:04 contained explicit full-index saved messages for 51 of them. A uniform draw without replacement selected 16 before inspecting their complexity:

```
random.Random(3329612950807171883).sample(sorted(eligible_ids), 16)
```

The seed was drawn once from OS randomness. Conditions matched the preceding benchmark: one connected theory at a time, t^{18} , `-w32`, no FORM-cache reuse, one observation per theory, and complete output written to a local file.

ID	Gauge group	TFORM s	Save s	Ratio %	Save/TFORM	GiB
19704	$SU(3) \times SU(4)$	9.00	142.23	6.33	15.80	3.55
19726	$SU(3) \times SU(5)$	14.96	147.70	10.13	9.87	5.49
19739	$SU(3) \times SU(5)$	14.96	76.92	19.45	5.14	5.49
19749	$SU(3) \times SU(5)$	14.90	105.61	14.11	7.09	5.50
19794	$SU(3) \times SU(6)$	9.50	270.95	3.51	28.52	3.65
19920	$SU(4) \times SU(4)$	9.23	251.74	3.67	27.27	3.56
20062	$SU(4) \times SU(5)$	9.48	701.20	1.35	73.97	3.66
20063	$SU(4) \times SU(5)$	9.65	251.11	3.84	26.02	3.66
20109	$SU(4) \times SU(6)$	9.01	304.90	2.96	33.84	3.56
20113	$SU(4) \times SU(6)$	9.17	117.40	7.81	12.80	3.55
20155	$SU(4) \times SU(6)$	9.38	106.25	8.83	11.33	3.65
20230	$SU(4) \times SU(7)$	15.07	148.71	10.13	9.87	5.50
20591	$SU(5) \times SU(6)$	14.95	122.91	12.16	8.22	5.49
20611	$SU(5) \times SU(6)$	9.33	91.99	10.14	9.86	3.61
20620	$SU(5) \times SU(6)$	15.10	180.11	8.38	11.93	5.47
22459	$SU(11) \times SU(11)$	15.22	645.29	2.36	42.40	5.50

Let T_{TFORM} be the isolated expansion wall time and T_{save} the historical interval from calculation start to the explicit full-index saved log line. The report uses unrounded times:

$$r = 100 T_{\text{TFORM}}/T_{\text{save}}, \quad q = T_{\text{save}}/T_{\text{TFORM}}.$$

All 16 succeeded: 9.00–15.22 s, total 188.91 s, median across theories 9.575 s; RSS 3.55–5.50 GiB. Here $r = 1.35$ – 19.45% and $q = 5.14$ – 73.97 . The later save interval includes the whole calculation and database save with 16 or 32 concurrent theories. The table compares different workloads and concurrency conditions; q is not a measured speedup of FORM or the complete index calculation.

Evidence. `output/benchmarks/tform32_recovered_random16_20260916/` retains the selection/population, original log lines, raw programs, output hashes, compressed outputs, measurements and ratio-bearing `report.md`. Neither benchmark was rerun for the 19 September documentation update.

6 Adams operations and LiE decomposition

6.1 The representation-ring calculation

For each irrep R and exponent vector $\mathbf{n} = (n_1, n_2, \dots)$, the cache stores the virtual decomposition

$$D_R(\mathbf{n}) = \text{Decomp} \left[\bigotimes_{j \geq 1} (\psi^j R)^{\otimes n_j} \right]. \quad (21)$$

An Adams operation scales every weight by j and can produce a virtual representation with negative coefficients [3, Secs. 3.4, 4.5]. For the A_1 fundamental,

$$\psi^2(\mathbf{2}) = \mathbf{3} - \mathbf{1}, \quad (22)$$

which LiE prints as a signed decomposition polynomial.

6.2 The LiE commands in plain language

LiE expression	Meaning
<code>Adams(2, [1], A1)</code>	Decomposes ψ^2 of the A_1 irrep with label [1]; the result is $-1X[0]+1X[2]$ [3, Sec. 4.5].
<code>tensor([1], [1], A1)</code>	Returns $1X[0]+1X[2]$ for $V_{[1]} \otimes V_{[1]}$ [3, Sec. 4.5].
<code>tensor(p, q, A1)</code>	Tensors two decomposition polynomials. The revised projector uses this for intermediate products only.
<code>tensor(p, q, [0], A1)</code>	LiE's targeted trivial coefficient operation. The revised projector instead performs the final dual-irrep pairing in Python.
<code>maxnodesN/maxobjectsN</code>	Raises LiE node/object limits. The code retries with these settings only after a failed ordinary run, avoiding eager large-allocation startup cost [3, Sec. 2.3].

A decomposition polynomial has the form

$$p = \sum_{\lambda \in P_+} m_\lambda X[\lambda_1, \dots, \lambda_r], \quad m_\lambda \in \mathbb{Z}. \quad (23)$$

The parser preserves the signed integer m_λ . It also removes LiE's optional “New tree space with maximum number of nodes” notification by recognizing the message prefix, rather than by deleting a fixed number of output characters.

7 Gauge-singlet projection for product groups

The gauge index is the Haar projection

$$\mathcal{I}(t, y, u) = \int_G d\mu_G(g) Z(t, y, u; g). \quad (24)$$

Equivalently, it is the multiplicity of the trivial representation in every coefficient. After FORM expansion, a typical formal-character monomial is

$$\mathcal{C} = \prod_p \prod_{j \geq 1} C_p(j)^{n_{p,j}}. \quad (25)$$

If $a(p)$ is the gauge factor associated with character p , define the virtual G_a representation

$$X_a(\mathcal{C}) = \bigotimes_{p: a(p)=a} \bigotimes_{j \geq 1} (\psi^j R_p)^{\otimes n_{p,j}}. \quad (26)$$

Because both the group and normalized Haar measure factorize,

$$d\mu_G = \prod_{a=1}^s d\mu_{G_a}, \quad (27)$$

$$\text{mult}_{\mathbf{1}_G}(\mathcal{C}) = \prod_{a=1}^s \text{mult}_{\mathbf{1}_a}(X_a(\mathcal{C})). \quad (28)$$

This is the precise rule implemented by `_project_terms`. It partitions a FORM term by gauge-factor position, batches distinct character-product singlet requests for each factor, and multiplies the returned integers. Empty character products have singlet multiplicity one. SQLite is checked before decomposition.

7.1 Exact pairing from character orthogonality

For compact G and normalized Haar measure,

$$\int_G \chi_\lambda(g) \overline{\chi_\mu(g)} d\mu(g) = \delta_{\lambda\mu}, \quad \chi_{\mu^*} = \overline{\chi_\mu}. \quad (28a)$$

Thus for two partial virtual characters,

$$A = \sum_\lambda a_\lambda \chi_\lambda, \quad B = \sum_\mu b_\mu \chi_\mu, \quad \boxed{[1](AB) = \sum_\lambda a_\lambda b_{\lambda^*}}. \quad (28b)$$

Character orthogonality is Corollary 35.9 of [6]. Equation (28b) follows by substitution and linearity, including negative coefficients. Equivalently, $(V \otimes W)^G \simeq \text{Hom}_G(V^*, W)$; the latter is the space of linear maps commuting with the group action. Schur's lemma selects dual irreps. The singlet functional here is bilinear; an ordinary coefficient dot product would incorrectly pair complex irreps with themselves.

7.2 Split before decomposition

A symbolic input $(\text{labels}, (n_1, n_2, \dots))$ stands for $\prod_j \psi^j(R)^{n_j}$. The revised procedure is:

1. Remove trivial-character factors, since $\psi^j(\mathbf{1}) = \mathbf{1}$.
2. Expand exponents into individual Adams factors and split them into two approximately balanced parts A and B .
3. Fully decompose the needed *partial* products, reusing persistent and in-memory entries. Individual Adams operations remain indivisible.
4. Apply (28b) using sparse dictionary lookups and save the resulting integer.

The splitter's cost estimate for an atom is

$$c(\lambda, j) = j \max \left(1, \sum_i \lambda_i \right). \quad (28c)$$

It places larger estimates first on the currently lighter side, preferring an already-present base irrep when the two costs tie. This source-derived heuristic balances approximate highest-weight sizes, not exact tensor complexity, and carries no optimality guarantee. Repeated factors of one base irrep can be split across the two sides before their complete product is decomposed.

For example, $\psi^4(\text{adj})\psi^5(\text{adj})$ needs only the two individual Adams expansions and their pairing. In contrast, $\psi^3(\text{adj})^3$ leaves the intermediate $\psi^3(\text{adj})^2$, which can remain expensive. The full decomposition API is still available when explicitly requested; the singlet path avoids tensor-decomposing the complete product.

7.3 Dual labels and signed examples

The highest weight of the dual irrep is

$$\lambda^* = -w_0\lambda. \quad (28d)$$

The map permutes fundamental weights [6, Prop. 32.1]. `_dual_permutation` obtains that permutation with the existing Sage duality helper and caches it; `_singlet_pair` applies it to label tuples. See also Sage's `dual()` and `inner_product()` [7]. The pairing iterates over the smaller support, using $O(r \min(N_A, N_B))$ label operations. The costly step can still be constructing A or B .

For $SU(3)$, $(a, b)^* = (b, a)$, so $[\mathbf{1}](\mathbf{3} \otimes \overline{\mathbf{3}}) = 1$ but $[\mathbf{1}](\mathbf{3} \otimes \mathbf{3}) = 0$. Actual conjugate orientations must remain in cache keys, even where theory enumeration identifies conjugates. For $SU(2)$, $\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}$ implies $[\mathbf{1}](\mathbf{2}^{\otimes 4}) = 1^2 + 1^2 = 2$. Pseudoreality adds no sign to this pairing. Signed Adams coefficients are different: from (22), $[\mathbf{1}](\psi^2\mathbf{2}) = -1$, while $[\mathbf{1}](\psi^2\mathbf{2})^2 = (-1)^2 + 1^2 = 2$.

7.4 Disconnected gauge sectors and the factorization identity

A product gauge group is not necessarily a product of independent theories: matter charged under several factors couples those factors. For each nonzero hypermultiplet h , define its gauge support

$$S_h = \{a : R_{h,a} \neq 1\}.$$

`split_disconnected_sectors` builds an undirected NetworkX graph with one vertex per gauge factor. For a support $(a_1, \dots, a_r)$, it inserts the $r - 1$ edges $(a_1, a_2), \dots, (a_1, a_r)$. A star joins every vertex of that support by a path; a complete clique would give the same connected components. Overlapping supports join transitively. Zero-multiplicity matter is ignored and isolated gauge vertices are retained. Connected components are obtained with `networkx.connected_components` [12].

For components $C_1, \dots, C_s$, each charged hyper belongs to exactly one component. The code preserves factor order, restricts representation and beta contribution dictionaries, and converts a one-factor product hyper into `HyperData`. Multi-factor components retain `ProductHyperData`. No discarded spectator singlet contributes an extra dimension factor.

Public API. `split_disconnected_sectors(factors, hypermultiplets)` returns a list of pairs: a tuple of `GaugeFactorData` and a list of `HyperData` or `ProductHyperData`. Input factor IDs and order are retained. The basic property calculator materializes these IDs as a tuple of tuples, not lazy iterators; it returns `None` for a non-SCFT candidate. The free sector contributes an empty tuple, serialized as an empty JSON array. Database ownership of those IDs is described in the implementation reference.

Let L_α denote the letters of component α and L_0 all matter with empty gauge support. The exponent splits because Adams substitution and summation are linear, so the plethystic identity gives

$$L = L_0 + \sum_{\alpha} L_{\alpha}, \quad \text{PE}[L] = \text{PE}[L_0] \prod_{\alpha} \text{PE}[L_{\alpha}].$$

Normalized Haar integration also factorizes, since each L_{α} depends only on $g_{\alpha} \in G_{C_{\alpha}}$:

$$\begin{aligned} \mathcal{I} &= \int \prod_{\alpha} d\mu_{\alpha}(g_{\alpha}) \text{PE}[L_0] \prod_{\alpha} \text{PE}[L_{\alpha}(g_{\alpha})] \\ &= \mathcal{I}_0 \prod_{\alpha} \mathcal{I}_{\alpha}, \quad \mathcal{I}_0 = \text{PE}[L_0], \quad \mathcal{I}_{\alpha} = \int d\mu_{\alpha} \text{PE}[L_{\alpha}]. \end{aligned}$$

This follows directly from the plethystic definition and product measure [2, 1]; the graph is the implementation’s test for these independent integration variables.

Why all-gauge-singlet matter appears once. A gauge-singlet hyper has empty support. Dropping it loses a physical free-matter factor. Copying it into each of s components would replace $\mathcal{I}_0$ by $\mathcal{I}_0^s$. The code puts all such hypers in one additional sector with no gauge factors, preserving each original multiplicity. “Once” means one assignment of each input species, not changing its multiplicity to one.

7.5 Exact multiplication, flavor characters and database reuse

At finite cutoff, multiplication and truncation commute in the needed sense:

$$\left[\prod_{\alpha=0}^s \mathcal{I}_{\alpha} \right]_N = \left[\prod_{\alpha=0}^s [\mathcal{I}_{\alpha}]_N \right]_N .$$

All sector indices have nonnegative t degree. Discarded powers above N cannot multiply later terms back down to N . Laurent powers of y, u do not alter that argument. `_truncate_index` filters the exact Sage monomial dictionary inclusively after each sector and each product. An identical sector within the same call is looked up or calculated once, but its index is still multiplied once per occurrence.

Flavor refinement. The same identity holds with external flavor variables x : $\mathcal{I}(t, y, u; x) = \mathcal{I}_0(t, y, u; x) \prod_{\alpha} \mathcal{I}_{\alpha}(t, y, u; x)$. Shared flavor variables are coefficients, not integrated gauge variables. Products of their characters describe tensor products and need not be decomposed first. For example, a free full hyper with $U(1)$ fugacity x contributes $\text{PE}[f_H(x + x^{-1})]$ once. Gauging a shared flavor subgroup introduces a common Haar integral and can connect sectors; one cannot then multiply separately projected indices. The current code is flavor-unrefined: this is the extension’s mathematical contract, not an implemented flavor API.

Optional theory database. `calculate_index` and `calculate_index_internal` accept `theory_db_connection=None`. For each gauged sector, `common.n2_theory_db.find_superconformal_index` matches the existing canonical Lagrangian hash and reads an index only if it is a JSON string with recorded `superconformal_index_order>=order`. Canonicalization retains the established full/half and conjugation conventions; factor IDs are ignored but input factor order remains significant. Missing/null indices and unknown/lower cutoffs are misses. A malformed index or one with negative t powers is rejected by the caller and recalculated. The cutoff is never inferred from the highest surviving monomial.

Misses use `_calculate_sector_index`: character basis, degree-bounded FORM expansion/cache and the existing LiE singlet projection. Free sectors bypass the theory lookup. No sector theories or indices are inserted. A supplied connection is borrowed read-only and is never committed, rolled back or closed by this calculation; it must belong to the calling process/thread and already have the current schema. GUI/database workers pass their own connection. Without one, sector splitting and multiplication still apply.

The character-cache file is a distinct parameter: `char_cache_database_path=None`. It replaces `cache_directory` and `database_path` in the index API; the index and database-worker CLI use `--char-cache-database`. The independent FORM file uses `form_cache_database_path`. The standalone character-cache class/builder retain their own directory/file API and CLI. Saved GUI key `cache/character_database` is unchanged.

7.6 Product-group bottlenecks and recorded evidence

The 14 September investigation found 39 full-index FORM timeouts in a 32-worker run through t^{18} : 29 inputs had disconnected gauge sectors. The main isolated cost was constructing the joint formal-character exponential, before Python parsing, LiE projection or database writes. A printing-disabled control for disconnected $SU(3) \times SU(3)$ theory 9780 still took 516.16 s and retained 2,649,141 formal terms, whereas the final projected index has 168 monomials. This locates that delay in symbolic computation, not an observed database deadlock. Large expansions also incur parsing and memory costs. The report’s saved timeout setting was 60 s; the code default was 600 s. Those are dated observations, not assertions about the user’s current settings.

Implementation	Effect and evidence
Disconnected-sector split	Projects small independent sectors before multiplying their final polynomials. On 15 September, theory 9780 took 13.753 s with one unique sector calculation and exact agreement with all 168 saved combined-result monomials. Character cache was a private populated copy; FORM cache was empty; no MySQL connection was used.
Degree-bounded FORM	Helps connected sectors too, by omitting products above the remaining degree budget. The 16 September matched bifundamental run is recorded in the FORM explanation.
Remaining limits	Connected sectors can still have many formal monomials and expensive LiE intermediates. Concurrent misses can still duplicate one FORM program; no in-progress job claim or streaming projection was added.

Evidence files. `output/benchmarks/product_form_20260914/report.md` retains the original profiling controls and inputs; `output/benchmarks/disconnected_sectors_20260915.json` records the implemented split. The earlier diagnostic degree-bounded variant became the production generator on 16 September. Historical timings with different printing or cache conditions are not interchangeable benchmarks.

Regression coverage. `test/test_disconnected_index.py` tests transitive and trifundamental connectivity, isolated vectors, singlet matter, zero multiplicities, factor order, repeated sectors and exact comparison with an unsplit calculation. Database cases cover missing/null indices, unknown, lower, equal and higher cutoffs, canonical identity and borrowed-connection ownership. The recorded 15 September combined suite passed 353 tests in 58.615 s with no skips. The subsequent cache-argument revision passed 355 tests in 60.129 s. These are implementation-time results, not reruns performed for this document update.

8 SQLite caches and complete Adams-cache builder

For the exponent vector in Eq. (21), define its total Adams order

$$q(\mathbf{n}) = \sum_{j \geq 1} j n_j. \quad (29)$$

The default local file is `<project-root>/char_decomposition_cache.db`. Python's `sqlite3` stores both full character decompositions and final singlet integers. The schema has `user_version=1`, independently of the MySQL theory database. There is no combined product-group decomposition table; identical simple factors reuse the same entries.

Table	Columns and primary key
<code>character_decompositions</code>	<code>algebra</code> TEXT, <code>dynkin_labels</code> TEXT, <code>adams_powers</code> TEXT, <code>adams_order</code> INTEGER, <code>terms_json</code> TEXT. Key: <code>(algebra, dynkin_labels, adams_powers)</code> .
<code>singlet_coefficients</code>	<code>algebra</code> TEXT, <code>product_key</code> TEXT, <code>coefficient</code> TEXT. Key: <code>(algebra, product_key)</code> .

All columns are non-null, and both tables use `WITHOUT ROWID`. Label and power tuples are compact JSON text; signed coefficients are decimal strings, avoiding SQLite's 64-bit integer limit [9]. Decomposition payloads are lists of `[labels, "coefficient"]` pairs. The full power tuple is required: total Adams order alone does not distinguish, for example, $R^{\otimes 2}$ from $\psi^2(R)$.

Symbolic singlet keys contain `["characters", canonical_product]`. Repeated base labels merge by adding exponents; factors sort in descending order. Already-decomposed inputs use a separate `"decompositions"` namespace. Zeros and negative coefficients are valid cache hits. Returned decompositions are copies, so caller mutation does not change a cached entry.

8.1 Lookup order and concurrency

A singlet lookup checks thread-local memory, then SQLite, before requesting any partial decomposition. Same-base-irrep partial products use persistent entries; mixed partial decompositions live only in the thread-local `partial_products` dictionary. Only the final complete-product scalar is persisted as a singlet.

Each thread owns a connection. PID checks avoid inherited connections; before starting a process pool the parent closes its connection. Workers calculate independent requests, and the parent batches completed writes by dependency level. Sequential batches flush after 64 entries and at completion. LiE work occurs outside write transactions. Failed calculations do not cache a guessed zero.

WAL allows concurrent readers and a writer, while writers remain serialized; it assumes a local filesystem [8]. The code uses a 30-second busy timeout and bounded retries. Python connection contexts manage transactions; this cache's context manager additionally closes the connection and clears thread-local entries [10].

Legacy JSON file discovery/import and `cache_path()` are removed. The new projection preserves existing SQLite keys and schema; no migration is needed. FORM expansion now has its own raw-program cache, described next.

8.2 Persisting the FORM expansion before projection

`index/form_expansion_cache.py` now owns `IndexFormTerm`, the exact parser, and `FormExpansionCache`. For raw program text P , the stored value is the parsed list of monomials before any gauge projection:

$$P \mapsto \{(c, \alpha, \beta, \gamma, ((p, \mathbf{n}_p))_p)\}, \quad ct^\alpha y^\beta u^\gamma \prod_{p,j} C_p(j)^{n_{p,j}}.$$

This notation is a direct description of the project record format. Each `IndexFormTerm` has a `Fraction` coefficient, integer fugacity powers, and a tuple of character indices and Adams-power tuples. The compact JSON row is `["numerator", "denominator", t_power, y_power, u_power, characters]`. `_encode_expansion` stores a list of these rows; `_decode_expansion` restores exact arbitrary-size fractions and nested tuples. No float conversion or representation decomposition occurs in this layer.

```
CREATE TABLE form_expansions (
  program TEXT COLLATE BINARY NOT NULL PRIMARY KEY,
  expansion_json TEXT NOT NULL
) WITHOUT ROWID;
```

The key is the complete program text, compared verbatim, not a hash. Whitespace, numeric multiplicities, symbol names and the truncation order matter. There is no program normalization or symbolic-multiplicity template. The default file is `form_expansion_cache.db`, separate from the character database; the FORM module does not set a separate `user_version`.

Lookup. `get_expansion(program)` reads and decodes a hit. A miss runs `run_form`, calls `parse_form_output`, encodes the terms and commits with `ONCONFLICT(program)DONOTHING`. Failed execution or parsing does not create an entry. Each thread opens its own connection, with a PID check before reusing it, WAL, a 30-second busy timeout and up to three attempts for busy/locked writes. FORM runs outside write transactions. Concurrent identical misses can execute FORM more than once, but retain only one complete row. The context manager closes the current thread's connection. These boundaries follow the implementation and SQLite's serialized-writer model [8, 10].

Index integration. `calculate_index` and `calculate_index_internal` accept `form_cache_database_path=`; the index CLI exposes `--form-cache-database`. Otherwise the FORM database is placed beside the selected character database. A hit bypasses FORM execution and FORM-output parsing, while group-specific singlet projection still runs. Orders below two return the vacuum without external execution or cache access.

Different theories can share this expansion when their raw programs are identical and they use the same FORM database. Group and irrep meanings are supplied afterward by each theory's character basis. Their final indices can therefore differ despite a shared expansion. This reuse applies to the supported non- SU groups too; it does not require a power-sum/Schur integration backend.

8.3 Large cache files and WAL lifetime

A several-gigabyte cache is not by itself a reason to replace SQLite [13]. For this implementation, distinguish indexed-key lookup from decoding one large JSON payload, scanning candidate keys for a dependency plan, and waiting for concurrent writes. Those costs need representative measurements; this session did not benchmark a several-gigabyte cache.

In WAL mode, `-wal` holds the write-ahead log and `-shm` coordinates access. Other open connections or an unclean process exit can leave these files after a program finishes. SQLite normally checkpoints and cleans up on the last clean close; their presence alone is not evidence of corruption [14]. Never remove a live WAL manually: committed data may still be there. A cache context manager closes its current thread's connection; that does not close another process's connection. No new checkpoint or worker-finalizer policy was implemented as part of this discussion.

8.4 Reusing lower-order character decompositions

Write the cached virtual character as

$$D_R(\mathbf{n}) = \sum_{\lambda} d_{\lambda} \chi_{\lambda} = \prod_{j \geq 1} (\psi^j \chi_R)^{n_j}, \quad q(\mathbf{n}) = \sum_j j n_j.$$

If $\mathbf{a} + \mathbf{b} = \mathbf{n}$ componentwise, multiplication in the representation ring gives

$$D_R(\mathbf{n}) = D_R(\mathbf{a}) D_R(\mathbf{b}), \quad d_{\nu} = \sum_{\lambda, \mu} a_{\lambda} b_{\mu} N_{\lambda\mu}^{\nu}, \quad \chi_{\lambda} \chi_{\mu} = \sum_{\nu} N_{\lambda\mu}^{\nu} \chi_{\nu}.$$

The identity follows by multiplying the defining Adams products; LiE's `tensor` implements the last decomposition [3, Sec. 4.5]. Signed coefficients remain valid. The two exact base cases are $D_R(\mathbf{e}_1) = \chi_R$ and $D_1(\mathbf{n}) = 1$; neither needs a LiE call.

`CharacterDecompositionCache._decomposition_dependencies` first tries the usual split: remove one factor at the smallest occupied Adams index. For at most two factors, or when both usual dependencies are available, it retains that split. Otherwise it searches lower-weighted-order keys for the *same* Cartan type and Dynkin labels, including thread-local entries and the known first Adams operation. It considers pairs already available whose padded exponent vectors sum to the requested vector.

Among alternative pairs, the score is the product of the two numbers of irreducible terms, with zero cost for a zero or scalar-trivial factor. Ties use the total number of terms and then the vectors themselves. This estimates tensor work; it does not predict the fastest LiE call or find a globally optimal factorization. The batch dependency planner and the actual calculation use the same selection. Existing SQLite keys, schema and thread/process ownership are unchanged.

For example, if $R^{\otimes 2}$ and $(\psi^2 R)^{\otimes 2}$ are stored, their product can be obtained with one tensor operation on those entries. Atomic prerequisites need not be recalculated just to reconstruct a cached composite. On-demand generation still constructs missing prerequisites when no suitable cached split is available. It does not automatically enumerate all lower orders.

Scope. This optimization is separate from the singlet splitter. The singlet path still decomposes two partial products and contracts dual labels; the new dependency choice helps construct the required same-irrep partial products. Intermediate full tensor decompositions can remain expensive even when all individual Adams operations have already been cached.

8.5 Building every product through a chosen Adams order

`build_decomposition_cache(cartan_type, dynkin_labels, max_adams_order, ...)` in `index/char_decomposition_cache.py` fills the existing decomposition table for one base irrep. At weighted order q it calls `common.math_utils.frobenius_solve(range(1, q+1), q)` to enumerate

$$\mathcal{P}_q = \left\{ (n_1, \dots, n_q) \in \mathbb{Z}_{\geq 0}^q : \sum_{j=1}^q j n_j = q \right\}.$$

Each solution specifies one Adams product. This is also the multiplicity description of an integer partition: $|\mathcal{P}_q| = p(q)$, with counts 1, 2, 3, 5, 7, 11 at orders one through six (29 rows in total).

Orders run sequentially. The builder reads existing keys at the current order, skips complete rows without decoding their payloads, and divides missing solutions into batches for a persistent `ProcessPoolExecutor` using `spawn`. Batches contain at most 64 requests, with several batches per worker to balance uneven costs. Workers read prerequisites and calculate; the parent commits each returned batch. It submits the next order only after the current one is fully committed. LiE never runs in a write transaction. Completed batches survive a later failure or interruption, so a rerun resumes from the missing entries. Independent builders may duplicate simultaneous misses; uniqueness and short transactions preserve complete stored rows.

Every composite at order q has a split into two strictly lower orders. Consequently both factors are already stored and at most one tensor operation is needed for that request, with scalar/zero shortcuts as usual. For an initially empty cache of a nontrivial irrep, one uninterrupted build needs only one individual Adams calculation per $j = 2, \dots, M$: $M - 1$ in total. This counts logical Adams calculations, not backend retries or concurrent duplicate builds. The builder fills same-irrep decompositions, not singlet coefficients, mixed-irrep products or the FORM database.

Cutoff for the full index. A letter starts at t^2 , and its j -th Adams contribution starts no earlier than t^{2j} . For each base irrep in a fixed gauge factor, its product therefore satisfies

$$\deg_t \geq 2 \sum_j j n_j = 2q \implies q \leq \left\lfloor \frac{N}{2} \right\rfloor \quad \text{for an index through } t^N.$$

This is a bound derived from the implemented single-letter functions. Use $M = \lfloor N/2 \rfloor$ for each required adjoint and matter irrep, retaining distinct conjugate labels. It is a sufficient bound, not a claim that every partition will occur in the projector. For product groups it holds factorwise; one bifundamental letter contributes a character to multiple factors, so one must not sum their separate orders and apply the same bound to that sum. Through t^{18} , $M = 9$ suffices. For $N < 2$ there is no cache to prebuild.

8.6 Builder API, dependencies and command-line examples

With Sage's environment active, prepare the A_2 fundamental through order nine:

```
sage -python -m index.char_decomposition_cache A2 \
  --dynkin-labels 1 0 --max-adams-order 9 --processes 4 \
  --cache-database /tmp/index-demo/char_decomposition_cache.db
```

Repeat for every required irrep. For A_2 SQCD these include $(0,1)$ and the adjoint $(1,1)$ in addition to $(1,0)$. The group rank does not multiply the maximum Adams order. The direct script entry point `sage-python index/char_decomposition_cache.py...` is also supported.

```
from index.char_decomposition_cache import build_decomposition_cache

if __name__ == "__main__":
    totals = build_decomposition_cache(
        "A2", (1, 0), 9, processes=4,
        database_path="/tmp/index-demo/char_decomposition_cache.db",
        progress=lambda q, total, computed:
            print(q, total, computed),
    )
```

The return value is `dict[int,int]` mapping each order to its total product count, including reused rows. The optional callback receives `(order,total,computed)` in the parent after that order is committed. `processes=1` is serial; the default is the available CPU count. The pool starts only when there is parallel work. Use a `__main__` guard in scripts with multiple processes.

The maximum order and worker count must be positive integers. Cartan type, rank and Dynkin labels are validated. `cache_directory=` and `database_path=` are mutually exclusive. Other keywords are `lie_executable=` and `timeout=` (600 seconds per LiE call by default; Python accepts `None` for no timeout). The CLI provides `--max-order` as an alias for `--max-adams-order`, as well as `--cache-directory`, `--lie-executable`, and numeric `--timeout`. It prints `computed/reused` counts and exits with status 2 on a reported failure.

The builder imports `frobenius_solve` lazily and therefore needs OR-Tools only when called; ordinary index/cache lookups do not acquire that dependency. Its enumeration and complete decompositions can grow much faster than the subset used by one index. Prebuilding is optional, and a fully warm repeat still enumerates solutions and checks keys even though it runs no LiE or worker pool.

Use the prepared character database and a shared FORM database for index calls:

```
sage -python -m index.n2_theory_index theory.json --order 18 \
  --char-cache-database /tmp/index-demo/char_decomposition_cache.db \
  --form-cache-database /tmp/index-demo/form_expansion_cache.db
```

Without the last option, the FORM database defaults to that same directory. The corresponding index Python keywords are `char_cache_database_path=` and `form_cache_database_path=`.

9 Python functions and their mathematical roles

Function or class	Mathematical/computational role
<code>_parse_input</code>	Resolves the ordered factors in Eq. (1) and validated hypermultiplet data.
<code>_matter_character_multiplicities</code>	Constructs Eq. (10), omits trivial factor characters, and applies conjugation as in Eq. (11).
<code>_character_basis</code>	Assigns each pair (a, R) one FORM function C_p and records vector and matter character monomials.
<code>_build_form_program</code>	Implements Eqs. (7), (17), (14), and (20), with the degree-bounded loop in Section 5.4.
<code>run_form</code>	Shared helper in <code>common/form_utils.py</code> ; runs FORM/TFORM in an isolated temporary directory and requires a successful, clean process result.
<code>FormExpansionCache. get_expansion</code>	Retrieves parsed terms by raw program; executes, parses and stores a miss.
<code>FormExpansionCache. parse_form_output</code>	Converts exact FORM monomials to <code>IndexFormTerm</code> records with <code>Fraction</code> coefficients.
<code>get_decomposition, get_decompositions</code>	Retrieve or construct Eq. (21); prefer available cached factors and batch missing prerequisites by tensor-factor count.
<code>build_decomposition_cache</code>	Enumerate all products at each weighted order; commit lower orders before parallel generation of the next.
<code>get_singlet_multiplicities</code>	Look up symbolic-product singlets first; calculate missing integers with <code>_character_singlets</code> .
<code>singlet_multiplicities</code>	Cache singlets from already-decomposed inputs; misses call <code>_compute_singlet_multiplicities</code> .
<code>_split_character_product</code>	Greedily split individual Adams factors using Eq. (28c). Repeated factors may separate.
<code>_character_singlets</code>	Internal <code>plan()</code> identifies needed parts; <code>expand()</code> builds and memoizes mixed parts before final pairing.
<code>_dual_permutation, _singlet_pair</code>	Cache the label permutation in Eq. (28d) and evaluate Eq. (28b), including signed coefficients.
<code>_tensor_decompositions</code>	LiE decomposition of intermediate products; zero and scalar-trivial factors are handled directly.
<code>_compute_singlet_multiplicities</code>	For decomposed inputs, sort by support size, recursively build two parts and pair dual labels.
<code>_project_terms</code>	Multiplies factorwise singlet multiplicities and collects equal powers of t, y, u .
<code>_to_sage_polynomial</code>	Returns an exact flat Laurent polynomial in $\mathbb{Q}[t^{\pm 1}, y^{\pm 1}, u^{\pm 1}]$; performs no representation calculation.
<code>split_disconnected_sectors</code>	Public graph partition; returns factor tuples and hyper lists, retaining one free sector.
<code>calculate_index</code>	Parses input, then delegates sector partition, optional stored-index reuse and truncated multiplication to <code>calculate_index_internal</code> (Section 7.4).
<code>calculate_index_from_file</code>	Loads JSON and delegates to <code>calculate_index</code> .

10 JSON examples and command-line use

A simple-factor theory retains the existing form:

```
{
  "algebra": "A2",
  "hypermultiplets": [
    {"representation": "fundamental", "number": 6, "kind": "full"}
  ]
}
```

A product theory names every gauge factor. Missing entries in `representations` mean singlets:

```
{
  "gauge_groups": [
    {"id": "left", "algebra": "A1"},
    {"id": "right", "algebra": "A1"}
  ],
  "hypermultiplets": [{
    "representations": {
      "left": "fundamental", "right": "fundamental"
    },
    "number": 2, "kind": "full"
  }]
}
```

With the Sage environment active and `form` and `lie` on `PATH`:

```
python -m index.n2_theory_index theory.json --order 10
```

The default files are `char_decomposition_cache.db` and `form_expansion_cache.db` at the project root. Use `--char-cache-database /path/to/cache.db` to select the character database file. The corresponding Python argument is `char_cache_database_path=`. The FORM database follows the character database directory unless `--form-cache-database` (or Python `form_cache_database_path=`) overrides it.

For example, an unlimited single-worker calculation through t^{18} is

```
result = calculate_index(
    data, 18, processes=1, timeout=None,
    char_cache_database_path="/tmp/fresh-index/characters.db",
    form_cache_database_path="/tmp/fresh-index/form.db",
)
```

Use previously nonexistent files for both databases for a fully cold-cache timing. Filling only the character database leaves FORM cold, and vice versa. `timeout=None` is passed through to FORM and LiE; it removes their subprocess timeout.

11 Validation and representative performance

11.1 Earlier backend replacement checks (1 September snapshot)

The former sparse Sage weight-lattice implementation was retained long enough to serve as an independent reference before replacement. Exact comparisons gave:

Theory	Order	Check
$SU(2)$ with four full fundamentals	t^{18}	exact agreement
$SU(3)$ with six full fundamentals	t^{18}	exact agreement
$SU(4)$ with eight full fundamentals	t^{18}	exact agreement
$SU(2) \times SU(2)$ bifundamental theory	t^{10}	independent exact agreement
Decoupled product theories	t^6	factorization identity
$SU(2)^3$ trifundamental half hyper	t^6	factor-order invariance

For conformal $SU(3)$ SQCD, the first nonzero terms are

$$\begin{aligned} \mathcal{I} = 1 + t^4 \left(\frac{36}{u^2} + u^4 \right) - t^5 u^2 (y + y^{-1}) \\ + t^6 \left(\frac{40}{u^3} - 36 + u^6 \right) + O(t^7). \end{aligned} \quad (30)$$

Representative product-group timings through t^{10} were:

Theory	Cold cache	Warm cache
$A_1 \times A_1$ bifundamental	1.030 s	0.042 s
$A_1 \times G_2$ mixed matter	0.274 s	0.127 s

These historical timings predate FORM caching: their warm runs remove repeated LiE work but still execute the FORM series expansion. They are machine-dependent. The complete project test run after that earlier replacement passed 55 tests; two live MySQL integration tests were skipped because no test database was requested.

11.2 SQLite and sparse-projection checks (9 September)

The temporary projection candidate matched all seven comparative index cases and 319 singlet queries against the previous SQLite implementation (a9b7009). Of these, 202 also matched independent $SU(2)/SU(3)$ weight calculations. All 216 dual-label comparisons across 18 Cartan types matched Sage. All 4,996 old SQLite singlet entries tested were reused without recalculation. These checks preceded promotion to 0a402ec, before the later FORM/planner/builder changes.

The full post-projection suite ran 170 tests in 11.461 s: 168 passed and two live MySQL tests were skipped. Seven new regressions cover signed complex/pseudoreal pairing, grouped Adams splitting, mixed products, independent weights, parallel generation, trivial factors and zeros. These are local validation results; they do not establish correctness at all ranks and cutoffs.

11.3 Earlier speedups and A32 profile, before FORM caching

Cold calculation	Previous (s)	Revised (s)
A_8, t^{18} , singlet projection	4.599	0.849
A_{16}, t^{12} , singlet projection	1.473	0.207
A_{32}, t^8 , singlet projection	0.469	0.117
A_8, t^{18} , complete index	6.213	2.443

These are three-trial medians with fresh cache files and one worker, for SQCD with $2(r+1)$ full fundamental hypers. Projection excludes the shared FORM expansion. Warm A8 complete-index time remained about 1.70 s. Four-worker comparisons also matched, though process startup can slow small cases.

On 10 September, $A_{32} = SU(33)$ with 66 full fundamental hypers completed through t^{18} with no timeout and one worker:

Stage	Cold (s)	Warm (s)
Input/validation/algebra initialization	4.483	0.009
FORM expansion	1.357	1.406
FORM output parsing	0.212	0.239
Singlet projection, including cache work	88.229	0.127
Complete calculation	94.293	1.786

One LiE call decomposing $\psi^3(\text{adj})^2$ took 82.699 s, or 87.7% of the cold total. Its inputs each have 10 irrep terms and its output has 91. All other LiE calls together took 5.066 s. The warm run made no LiE calls. Both index results contain 117 nonzero monomials and match exactly. This is one wall-clock profile, not a median; the warm run shares Sage initialization but uses a new cache client.

Evidence and reproduction commands are in `output/benchmarks/singlet_projection_benchmark.md` and `output/benchmarks/a32_unlimited/report.md`. The unlimited run shows that the remaining cost is an intermediate full decomposition, not the final sparse pairing.

11.4 FORM cache measurements with character projection prefilled

The controlled comparison used nine theories at t^{12} and t^{18} , one warmup and five measured repetitions per mode, with fresh clients and one worker. Both decompositions and final singlet coefficients were already filled. The character database was query-only and guards prohibited LiE execution. All 270 timed indices agreed exactly. Medians below are full calculation times in seconds, excluding setup and Python/Sage startup.

Theory, through t^{18}	Disabled	Miss/save	Hit	Speedup
$SU(2)$, 4 fundamentals	0.4323	0.4916	0.0617	$7.00\times$
$SU(3)$, 6 fundamentals	1.7220	1.7959	0.2304	$7.48\times$
$SU(5)$, 10 fundamentals	1.7265	1.7839	0.2648	$6.52\times$
$SU(3)$, 1 adjoint ($\mathcal{N} = 4$)	0.2808	0.3180	0.0429	$6.54\times$
$Sp(2)$, 6 fundamentals	0.4434	0.4817	0.0634	$7.00\times$
$Spin(8)$, 6 vectors	0.4329	0.4848	0.0655	$6.60\times$
G_2 , 4 fundamentals	0.4418	0.4874	0.0640	$6.90\times$
$SU(2)^2$, 2 bifundamentals	2.7421	2.7843	0.3251	$8.44\times$
$SU(5)$, symmetric + antisymmetric	13.7565	14.1896	1.8771	$7.33\times$

All matter in this table is full hypermultiplets; $Sp(2) = USp(4)$. Disabled mode bypasses FORM SQLite and executes/parses FORM; miss/save begins with an initialized empty FORM database and includes serialization and insertion. Hit mode decodes an existing expansion with FORM execution forbidden. The speedup is disabled time divided by hit time. At t^{12} the range was $3.23\text{--}6.92\times$. These are warm-filesystem measurements on this machine, not cold-disk or contention benchmarks. Evidence: `report.md`, `results.json` and `benchmark.py` under `output/benchmarks/form_expansion_theories/`.

Cross-theory hits. Two pairs share byte-identical programs: $SU(2)$ with four fundamentals and G_2 with four fundamentals; $Sp(2)$ with six fundamentals and $Spin(8)$ with six vectors. Both directions at both orders gave eight passing reuse checks. The first theory executed FORM once; a new client for the second executed/parsed FORM zero times, decoded once, and left the database at one row. Each final index matched its own independent fresh-FORM result, while the paired final indices differed. A changed-program control missed as expected. See `cross_theory_reuse.md`, `cross_theory_reuse.json` and `check_cross_theory_reuse.py` in that same directory.

11.5 I/O cost, decomposition planning and builder verification

A separate I/O benchmark used the 113,831-term t^{18} expansion: a 589-byte program and 5,907,568-byte JSON payload (5.91 MB). Thirty measured repetitions followed three warmups, using SQLite WAL with synchronous FULL on the project filesystem. Medians are milliseconds:

Operation	Median (ms)
Serialize exact terms	179.32
SQLite insert and commit	42.07
Total save, excluding database initialization	221.51
SQLite read and fetch JSON	3.08
Reconstruct <code>IndexFormTerm</code> objects	866.42
Hit with an already open connection	870.81
Hit with a new client, including close	894.65

Initialization separately took 31.43 ms. No FORM execution/parsing is included; every round trip was checked exactly. Python reconstruction dominated loading. Individual component medians need not add to the median total. This is a single-process warm-filesystem test, not concurrent lock contention. Evidence: `output/benchmarks/form_expansion_cache_io/report.md`, `timings.json` and `benchmark.py`.

Cached-factor planner. The temporary candidate matched 130 products over A_1 , A_2 fundamental/adjoint, C_2 and G_2 in serial and three-worker runs; $SU(2)$ cases also matched an independent weight oracle. Prepared-cache misses improved $1.31\text{--}1.72\times$ for the single-request API and $1.03\text{--}1.17\times$ for the batch API, reducing two tensor calls to one. A case with only composite entries also reduced one Adams call to zero. However, 72 full-index comparisons over six theories, with FORM prefilled and either a cold character cache or a cache filled through t^{12} , showed *no consistent overall speedup*. All indices agreed and tensor-call counts were unchanged. The retained baseline, candidate, report, scripts and raw measurements are in `output/benchmarks/adams_cache_planner/`.

Complete builder. Twelve regressions cover all 29 $SU(2)$ products through order six against independent weights, serial/parallel agreement for A_1, A_2, C_2, G_2 through order five, pool reuse, warm skips, partial-cache resume, failure persistence, input validation and both CLI entry points. A real worker log records exactly one Adams call at each of orders two through six and 23 tensor calls, with multiple worker PIDs. This verifies the operation-count claim on the tested build; no broad prebuild speedup has been measured.

The full suite rerun during the 10 September 2026 cache-documentation update ran **209 tests: 207 passed and two live-MySQL tests skipped**. That run used actual Sage/FORM/LiE for backend tests; its MySQL migration and insertion unit tests used mocks unless a test server was explicitly configured. Schema migration tests were later removed when the production layout was reset to schema 1. The new modules are `test_form_expansion_cache.py`, `test_cached_decomposition_planning.py` and `test_build_decomposition_cache.py`. These measurements and algorithms are source-derived project evidence, not results attributed to external papers.

12 Interpretation and limitations

- The result is a signed protected index, not a complete operator spectrum.
- Flavor fugacities are unrefined. Matter multiplicities are integer coefficients.
- The code supports semisimple products of the listed finite simple algebras, but not $U(1)$ gauge factors.
- The Lie algebra does not determine the global form of the gauge group, discrete theta data, or the spectrum of line operators.
- Input validation checks representation syntax and physical admissibility of a half hyper. The index routine does not require the beta functions or global gauge anomalies to vanish; those are separate checker outputs.
- High orders still produce many FORM monomials and expensive intermediate LiE products. The cache reduces repeated work, and sparse pairing avoids the complete product's decomposition; neither removes all combinatorial growth or guarantees an optimal intermediate split.

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Earlier references were checked on 10 September; the added software references were checked on 16 September 2026. The splitting heuristic, function map and measurements are derived from the project source and local validation, rather than attributed to the representation-theory literature.